

PORTABLE



GRIP

Grounded Reasoning via Image Prompting
Medical VQA Pipeline | Chest X-Ray Analysis

Evaluating Visual Grounding in Multimodal Large Language Models **for** **Medical Question** **Answering**

Candidate

Riccardo Balzani

Supervisor

Prof. Gianluca Moro

Co-Supervisor

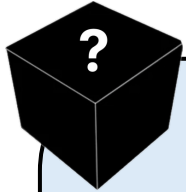
Dr. Lorenzo Molfetta

Co-Supervisor

Dr. Stefano Fantazzini

**Unibo
NLP**

Challenges in Medical VQA

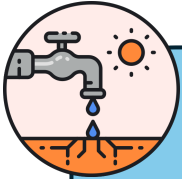


Black-box reasoning: End-to-end VLMs obscure *which anatomical regions* informed the diagnostic answer, undermining clinical trust



Visual hallucination: Models generate clinically plausible but spatially ungrounded findings based on textual biases rather than actual image content

Challenges in Medical VQA



Data scarcity:

Medical imaging data is scarce, heavily regulated, and expensive to annotate, leading to poor cross-domain generalisation



Domain-specific terminology: General-purpose NLP models fail to recognise clinical entities (e.g., *costophrenic angle*, *pneumothorax*, *air bronchograms*)



Evaluation limitations: Standard metrics like Exact Match rely on rigid string comparison and miss semantic equivalence (e.g., "CXR" vs. "Chest X-ray")

Purpose of the Thesis



Core question: Can explicit Visual Grounding of clinically relevant Regions of Interest (Rols) improve the accuracy and interpretability of Medical VQA?



Approach: Design a modular pipeline that **localises** pathological regions **before** answer generation, replacing black-box whole-image reasoning with focused, region-aware inference



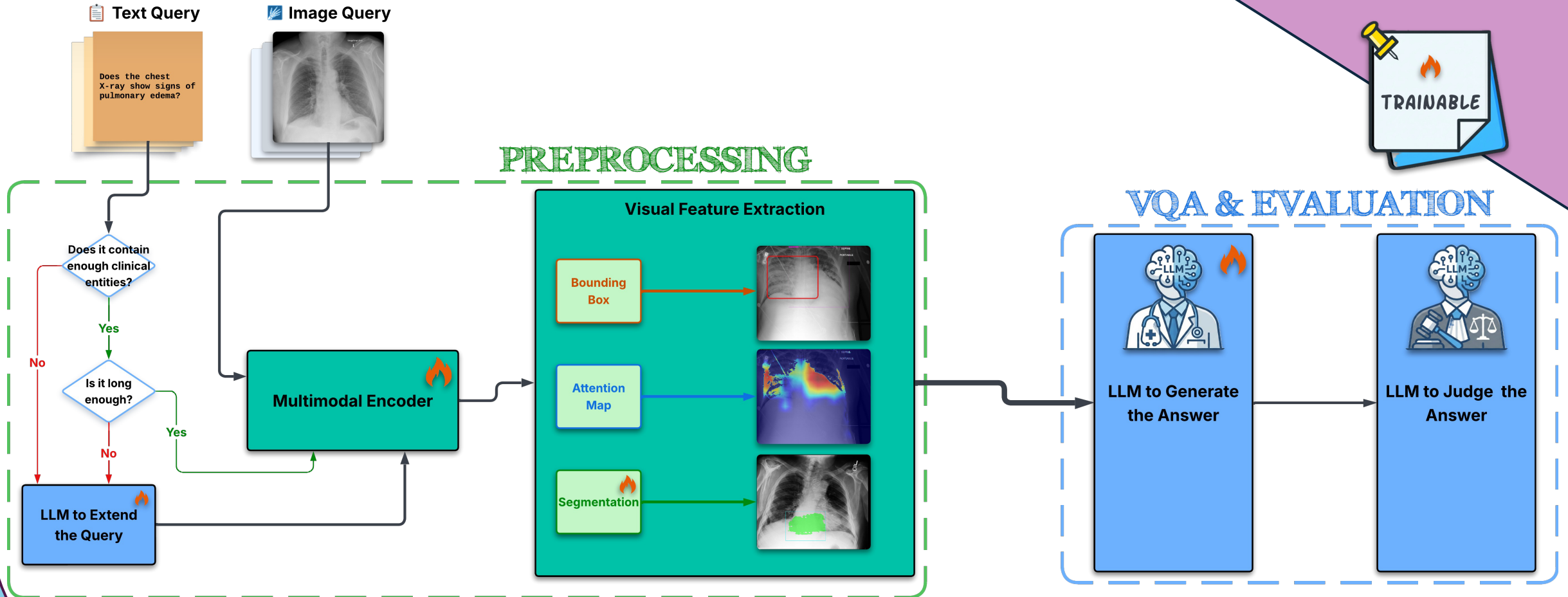
Key hypothesis: Showing the model **where to look** (via highlighted regions) produces better diagnostic answers than giving it the entire unprocessed image

GRIP - Grounded Reasoning via Image Prompting



GRIP

Grounded Reasoning via Image Prompting





GRIP

Grounded Reasoning via Image Prompting

Prompt Enhancement Layer

Text Query

Does the chest X-ray show signs of pulmonary edema?



Purpose: Ensure the clinical question is **specific enough** to drive meaningful visual attention

Does it contain enough clinical entities?

Yes

Is it long enough?

Yes

No

No

LLM to Extend the Query

Module: Biomedical NER pipeline trained on scientific corpora, recognising UMLS entity categories:

- Anatomical structures
- Pathological findings
- Medical
- Clinical descriptors

Decision criterion: A query is considered "clinically detailed" only if it contains at least 2 medical entities **and** at least 5 words

Routing logic:

- If the query is detailed enough → pass it through unchanged
- If the query is too brief or vague → expand it via a generative LLM into a richer clinical prompt



GRIP

Grounded Reasoning via Image Prompting

Visual Feature Extraction

Shared upstream signal: Attention map from BiomedCLIP + gScoreCAM

Bounding Box:

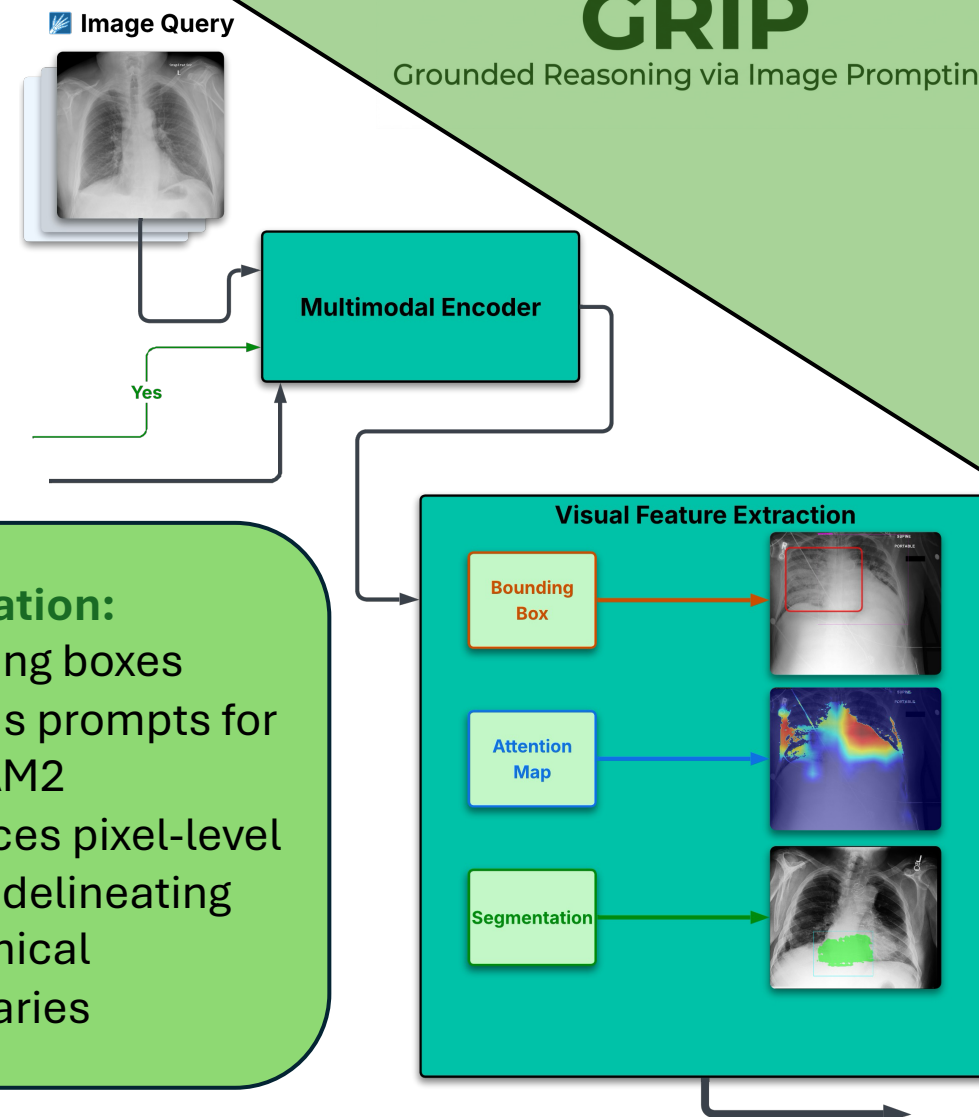
- Binarise the attention map via a threshold, extract contours, apply adaptive padding
- Draw coloured rectangles around detected regions on the original image

Heatmap Overlay:

- Map attention values to a colour scale
- Blend the colourmap with the original image
- Optional body mask suppresses non-anatomical regions

Segmentation:

- Bounding boxes serve as prompts for MedSAM2
- Produces pixel-level masks delineating anatomical boundaries

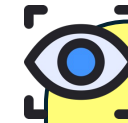
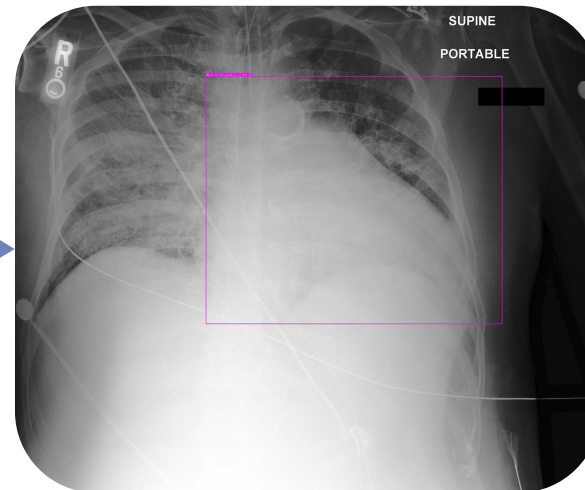
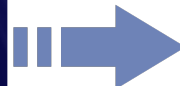
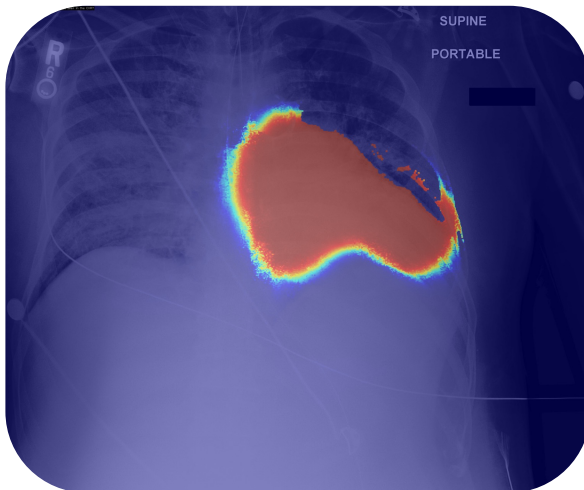
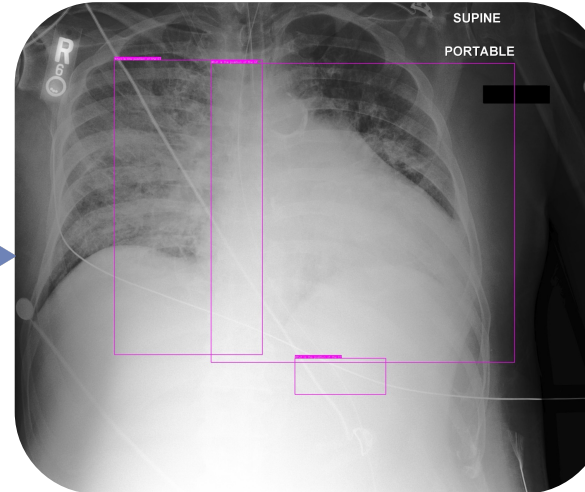
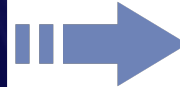
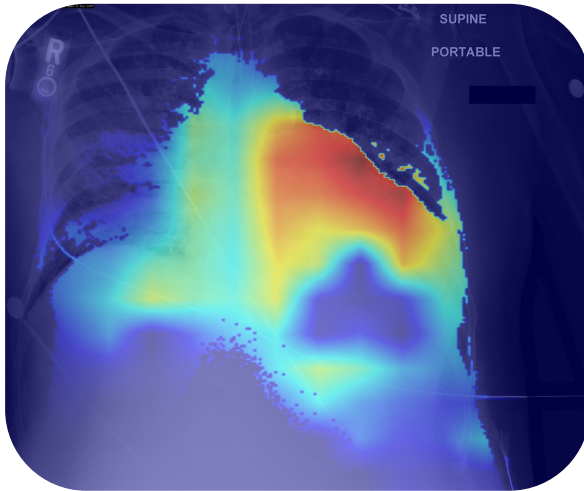
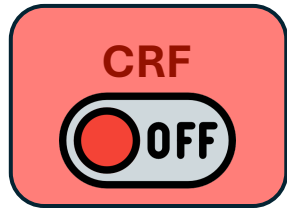


Insight: where is the model looking?



GRIP

Grounded Reasoning via Image Prompting



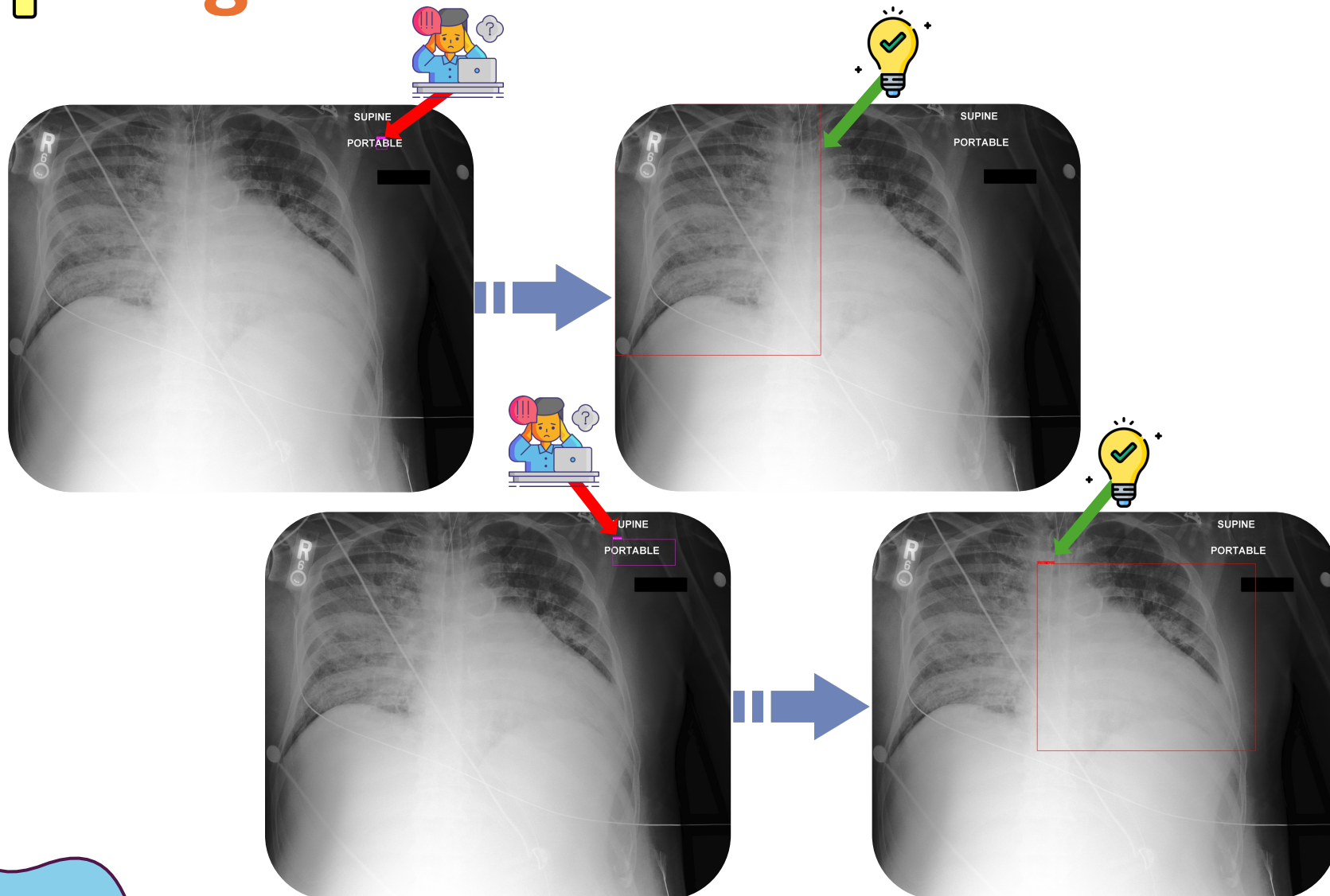
Attention Visualization using Grad-CAM, which generates a color-coded heatmap over the chest X-ray.



GRIP

Grounded Reasoning via Image Prompting

Insight: Anatomical Guardrails



Body Mask

Targets high-intensity artifacts. It creates a "spatial fence" around the gray-scale anatomy, preventing the model's focus from leaking into non-clinical margins or digital annotations.



Anatomical Check

Targets zero-signal regions. It automatically discards any detection generated outside the body.

Experimental Setup



GRIP

Grounded Reasoning via Image Prompting



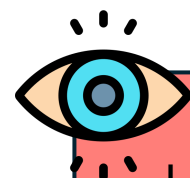
MedGemma 1.5 4B IT
Lightweight, biomedical-tuned



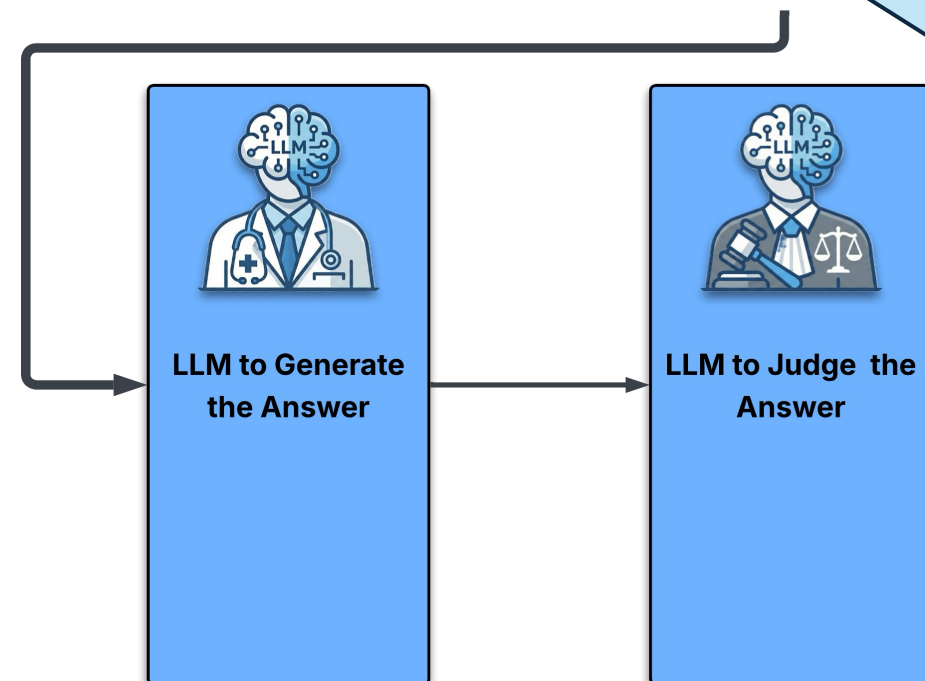
LLM-as-a-Judge
(Qwen2.5-7B-Instruct): An automated medical evaluator that scores answers on 3 criteria



MedGemma 4B IT
Same size, extended biomedical training corpus



OctoMed 7B
Larger model with built-in Chain-of-Thought reasoning



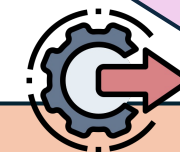
Evaluation Metrics



ROUGE-1 / ROUGE-L:
Measure word overlap
between prediction and
reference
Exact Match (EM):
Rigid string comparison
(correct only if identical)



1. **Semantic equivalence** (accepts medical synonyms and abbreviations)
2. **Medical specificity** (penalises missing laterality, severity, location)
3. **Hallucination penalisation** (flags unsupported medical claims)



OUTPUTS

Verdict

(Correct/Incorrect)

Confidence

(High/Med/Low)

Explanation

Datasets



MIMIC-CXR-Ext

- Derived from MIMIC-CXR database; 42M+ QA pairs from radiology reports
- QA pairs auto-generated via LLMs from scene graphs
- Stratified sampling preserving pathology distribution to handle class imbalance



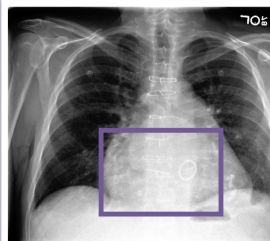
Role

Large-scale VQA evaluation

Q: Describe all abnormal findings in the given study.

There is severe cardiomegaly, predominantly right ventricular.

POSITIVE CERTAIN RIGHT



Categories:

ANATOMICAL FINDING DISEASE

Subcategories:

CARDIAC STRUCTURES CARDIOVASCULAR DISEASES

Findings:

cardiomegaly

Modifiers:

severity: severe

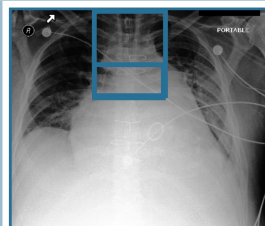
Regions:

heart right

Q: Status post intubation; the endotracheal tube should be evaluated.

The endotracheal tube is currently 6 cm from the carina and could be advanced 1-2 cm for optimal placement.

POSITIVE CERTAIN



Categories: Subcategories:

DEVICE DEVICE PART TUBES LINES PORTS

Findings:

endotracheal tube tip tube

Modifiers:

severity: severe

Regions:

chest trachea tracheal bifurcation

Datasets

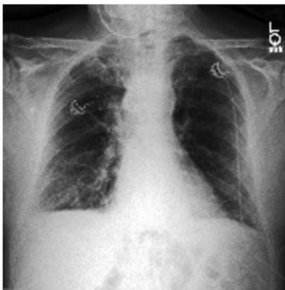
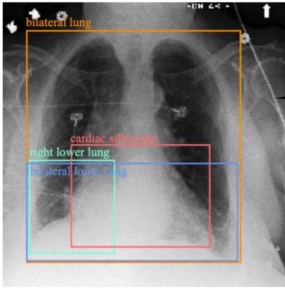


GEMeX VQA

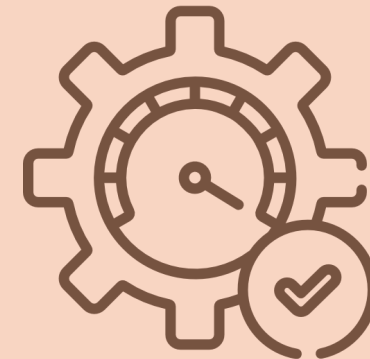
- **Expert-annotated bounding boxes** verified by board-certified radiologists
- **Multi-modal explainability:** visual grounding + textual rationales
- Diverse question taxonomy: finding identification, attribute recognition, abnormality detection
- Only ~50 tuples available; used as qualitative Gold Standard testbed

Role

Bounding box optimisation with clinically validated ground truth

Dataset	VQA Type	Example
 MIMIC-CXR-VQA	Closed-ended	Closed-ended VQA: Q: Is any devices present within the right atrium? A: No.
	Open-ended	Open-ended VQA: Q: What are all the diseases identifiable within the right hilar structures? A: A small right pleural effusion.
 GEMeX	Closed-ended	Multi-choice VQA: Q: Which regions on the X-ray show signs of abnormalities? C: ["A: Bilateral lungs", "B: Right costophrenic angle", "C: Bilateral lower lung", "D: Cardiac region"] A: ["A", "B", "C", "D"] Reason: Abnormalities are seen in the bilateral lungs (hyperinflation), right costophrenic angle (blunting), bilateral lower lung (atelectasis), and cardiac silhouette (enlargement). Bounding Box: [[28, 40, 167, 190], [28, 124, 85, 184], [28, 126, 165, 189], [57, 114, 147, 180]]
	Open-ended	
	Single-choice	
	Multi-choice	

Hyperparameter Optimisation



Objective: Find the best bounding box extraction settings; evaluated via IoU against GEMeX expert annotations



Exploration: prompting strategy, CAM threshold, CRF refinement, padding mode



Overall winner

- Question-driven
- threshold=0.30
- raw (no CRF),
- loose padding

IoU
0.2908

Results



Models		Pipeline	ROUGE Score			Accuracy	
Gen.	Judge	(Workflow Stages)	R-1	R-2	R-L	EM	Judge
MedGemma 1.5 4B IT	Qwen2.5 7B-Instr.	→	5.61	1.00	5.59	0.25	45.62
		→	11.49	1.14	11.44	3.80	47.40
		→	29.37	<u>1.83</u>	29.34	23.15	66.81
		→ →	<u>24.82</u>	1.65	<u>24.80</u>	<u>21.83</u>	37.26
		→	14.82	2.45	14.82	1.50	45.26
		→ →	11.76	1.07	11.74	5.80	<u>48.27</u>
MedGemma 4B IT	Qwen2.5 7B-Instr.	→	13.81	2.39	13.71	1.53	<u>46.73</u>
		→	29.52	1.89	29.45	14.20	46.85
		→	37.48	2.79	37.43	34.60	43.59
		→ →	24.12	4.05	24.07	13.53	41.31
		→	<u>32.23</u>	<u>2.89</u>	<u>32.14</u>	<u>31.15</u>	45.03
OctoMed 7B IT	Qwen2.5 7B-Instr.	→	30.31	1.47	30.24	24.43	45.43
		→	36.03	1.13	35.99	<u>29.20</u>	<u>45.70</u>
		→	34.54	1.72	34.50	29.40	43.90
		→ →	35.22	1.42	35.18	27.45	45.80
		→	<u>35.80</u>	<u>1.52</u>	<u>35.77</u>	29.12	44.99

Legend: = Query; = NER enabled; = Raw Image (Baseline); = Attention Map; = Bounding Box; = Segmentation; = Generation; = Evaluation.

Note: Gray-shaded cells indicate the **global** best performance across all models for a given metric. **Bold** values denote the best performance *locally* within each model's experiment class. Underlined values denote the second-best performance within each model's experiment class. Values are expressed in percentages.

Results



Model	Pipeline (Workflow Stages)	Exact Match		ROUGE Scores		
		Acc. (%)	Δ Baseline	R-1	R-2	R-L
MedGemma 1.5 4B IT	📄 → 🖼️	0.25	—	0.0561	0.0100	0.0559
	📄 → 🗺️	23.15	+22.90	0.2937	<u>0.0183</u>	0.2934
	📄 → 🗑️ → 🗺️	<u>21.83</u>	+21.58	<u>0.2482</u>	0.0165	<u>0.2480</u>
	📄 → 🗑️	3.80	+3.55	0.1149	0.0114	0.1144
	📄 → 🧩	1.50	+1.25	0.1482	0.0245	0.1482
	📄 → 🗑️ → 🧩	5.80	+5.55	0.1176	0.0107	0.1174
MedGemma 4B IT	📄 → 🖼️	1.53	—	0.1381	0.0239	0.1371
	📄 → 🗺️	34.60	+33.07	0.3748	0.0279	0.3743
	📄 → 🗑️	14.20	+12.67	0.2952	0.0189	0.2945
	📄 → 🧩	<u>31.15</u>	+29.62	<u>0.3223</u>	<u>0.0289</u>	<u>0.3214</u>
	📄 → 🗑️ → 🧩	13.53	+12.00	0.2412	0.0405	0.2407
OctoMed 7B	📄 → 🖼️	24.43	—	0.3031	0.0147	0.3024
	📄 → 🗺️	29.40	+4.97	0.3454	0.0172	0.3450
	📄 → 🗑️	<u>29.20</u>	+4.77	0.3603	0.0113	0.3599
	📄 → 🗑️ → 🗑️	27.45	+3.02	0.3522	0.0142	0.3518
	📄 → 🧩	29.12	+4.69	<u>0.3580</u>	<u>0.0152</u>	<u>0.3577</u>

Legend: 📄 = Query; 🗑️ = NER enabled; 🖼️ = Raw Image (Baseline); 🗺️ = Attention Map; 🗑️ = Bounding Box; 🧩 = Segmentation.

Note: The Δ Baseline column reports the absolute improvement in EM accuracy over the raw-image baseline for the same model. Gray-shaded cells indicate the **global** best performance across all models for a given metric. **Bold** values denote the best performance *locally* within each model's experiment class. Underlined values denote the second-best performance within each model's experiment class. ROUGE scores are reported on a 0–1 scale.

Results



Visual Grounding consistently outperforms baselines across all models:

- **MedGemma 1.5 4B IT:** ROUGE-1 from 5.61% to **29.37%** (5.2x increase) with Attention Map
- **MedGemma 4B IT:** ROUGE-1 from 13.81% to **37.48%** (2.7x); EM from 1.53% to **34.60%**
- **OctoMed 7B:** ROUGE-1 from 30.31% to **36.03%** (1.2x) with Bounding Box



Attention Map is the dominant prompting strategy, providing continuous, spatially dense cues



Visual Grounding narrows the model capacity gap:

Grounded MedGemma 1.5 (29.37%) approaches ungrounded OctoMed 7B baseline (30.31%)



21 percentage-point increase in diagnostic accuracy (EM) for the best configuration



NER routing: Model-dependent trade-off; beneficial for OctoMed, detrimental for smaller models



Thanks for the Attention

Which is “All You Need”

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

GRIP - Grounded Reasoning via Image Prompting



GRIP

Grounded Reasoning via Image Prompting

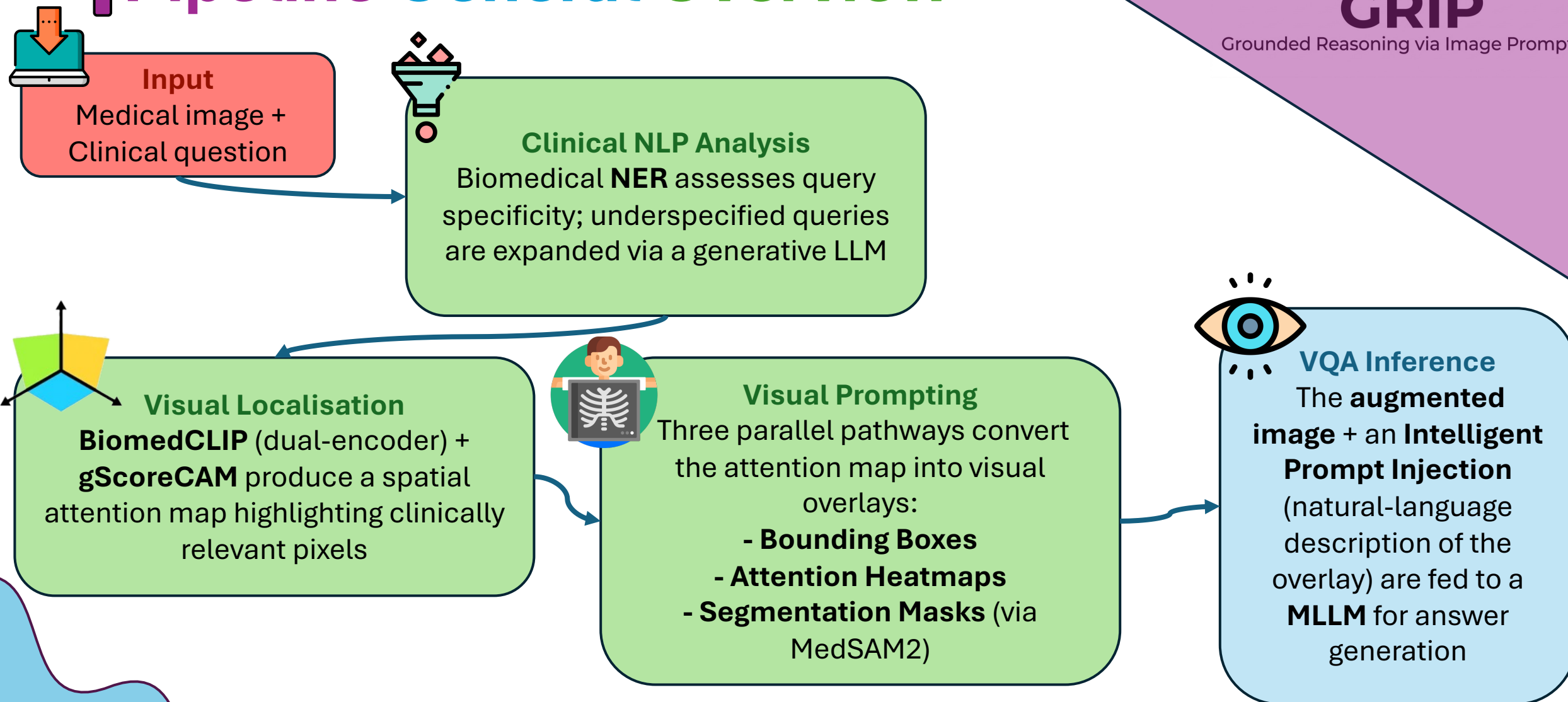
Medical VQA Pipeline | Chest X-Ray Analysis



GRIP

Grounded Reasoning via Image Prompting

Pipeline General Overview

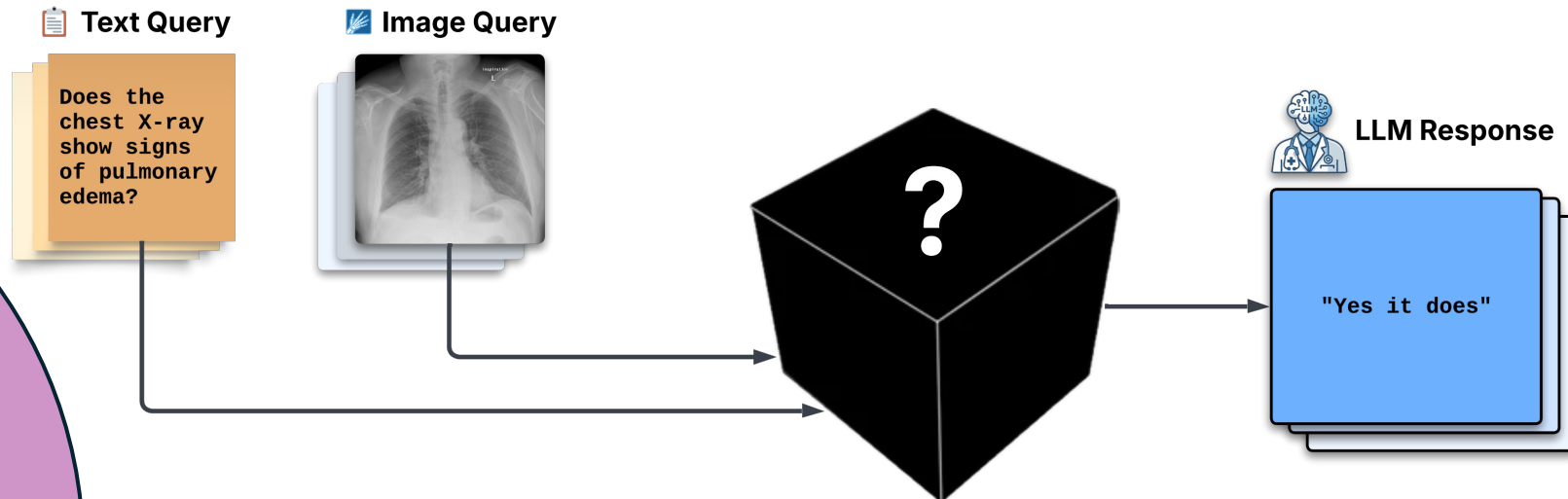


Challenges in Medical VQA



Black-box reasoning

End-to-end VLMs obscure *which anatomical regions* informed the diagnostic answer, undermining clinical trust





| GRIP - Grounded Reasoning via Image Prompting